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#include<iostream>
using namespace std;

float F(float x);
float trapzd(float a, float b, int n);

int main()
{
    int n;
    float a, b, s;
    cout << "\n a,b & n:";
    cin >> a >> b >> n;
    s = trapzd(a, b, n);
    cout << "\n Integral= " << s << endl;
    return 0;
}

float trapzd(float a, float b, int n)
{
    float x, tnm, s, sum, del;
    int j;
    del = (b - a) / n;
    s = 0.5 * del * (F(a) + F(b));
    sum = 0;
    if (n == 1)
    {
        return s;
    }
    else
    {
        for (j = 1; j < n ; j++)
        {
            x = a + j * del;
            sum += F(x);
        }
        s = s + del * sum;
        return s;
    }
}

float F(float x)
{
    float y;
    y = 1.0 / (1.0 + x + x * x + x * x * x);
    return y;
}

```